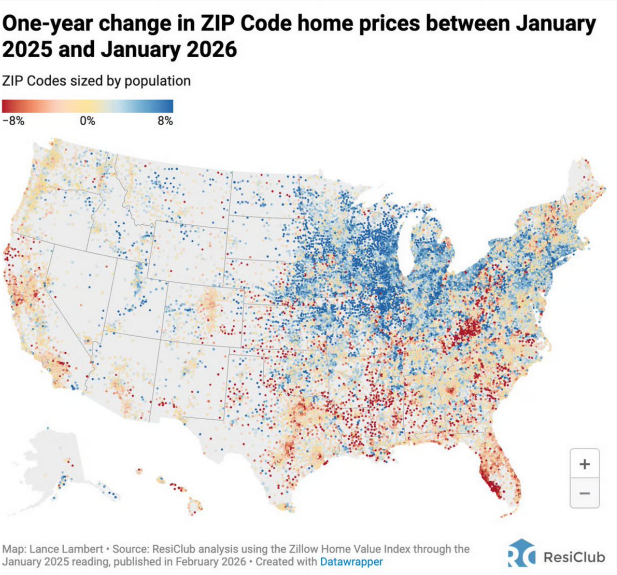
Does Regulatory Discretion Reduce Housing Supply? Evidence from Chicago's Aldermanic Privilege

Jacob Herbstman (University of Chicago)

March 2026

Motivation


One-year change in ZIP Code home prices between January 2025 and January 2026
ZIP Codes sized by population
−8%
0%
8%
Map: Lance Lambert • Source: ResiClub analysis using the Zillow Home Value Index through the January 2025 reading, published in February 2026 • Created with Datawrapper
ResiClub

Motivation

- Housing prices and rents are skyrocketing in many U.S. cities, including Chicago
  - Limited housing supply in high demand areas is widely accepted as the key driver of these price increases
  - Land use regulations and local control are often blamed for limiting supply
- Most research focuses on formal institutions such as by-right zoning laws
  - But regulatory discretion and stringency are also key determinants of housing supply and are less studied
- **Chicago**: Unique informal institution of *aldermanic privilege*
  - Offers a unique setting where de facto regulatory environment changes discretely at ward boundaries

What is Aldermanic Privilege?

- **City council logistics**
  - 50 wards + aldermen, each voted on only by the residents of their ward.
  - Each ward has a population of about 50,000 people.
  - 4 year terms, but aldermen are often in office for decades
- Core feature: **Deference**
  - Aldermanic privilege is not written in any law but is universally acknowledged
  - "Very few things happen that aldermen don't want to happen." (Zhang, 2011)
  - Often referred to as "little mayors" ruling wards as "fiefdoms"
- **Chicago gives more power to local politicians than many other U.S. cities**:
  - Houston has no explicit zoning code and centralized development approvals
  - Minneapolis has reduced councilmember discretion in recent years
  - Both cities have seen below-median rent and home price growth compared to peer cities (Hembre et al., 2025; Heckler, 2026)

Aldermanic Privilege is a Hotly Contested Issue in Chicago

CITY HALL NEWS ILLINOIS

Chicago wrongly limited affordable housing with aldermanic prerogative, HUD says

Feds say giving council members "a local veto over proposals to build affordable housing" has meant it's "rarely, if ever, constructed in the majority-white wards that have the least affordable housing."

By Brett Chase | Updated Nov 28, 2023, 5:54pm CDT

May 20, 2019

Mayor Lightfoot Signs First Executive Order to Limit Aldermanic Prerogative

In Her First Official Act as Mayor, Lightfoot Instructs Departments to Begin Process of Ending Aldermanic Prerogative as Part of Broader Ethics Reform

Mayor's Press Office 312.744.3334

Download this Press Release

POLITICS

Developer Moves Forward With Lincoln Park Apartment Complex, Setting Stage for Fight Over Aldermanic Prerogative

Heather Cherone | January 9, 2025, 5:00 am

Controversial Old Town Project With Proposed 44-Story Tower Rejected By Alderman

Ald. Brian Hopkins said he won't support Fern Hill's plan for North and LaSalle. But a new design could be coming.

By Patrick Filbin January 24, 2025 Updated January 27, 2025

Sterling Bay's Marcey Street Development Still Stalled After Heated City Council Exchange

The development would include more than 600 apartments in two towers in Lincoln Park near the Chicago River. Ald. Scott Waguespack opposes the plan in its current form.

By Quinn Myers December 11, 2024

This Paper

Research Question: Does the heterogeneity in regulatory stringency induced by the unique institution of aldermanic privilege reduce housing supply and raise prices?

Empirical strategy: Border discontinuity design at ward boundaries

- Look at multifamily development and rents near ward borders
- **Key assumption**: neighborhood characteristics vary smoothly across the border but the regulatory environment changes discretely

Preview of Results

1. More stringent aldermen $\Rightarrow$ lowers multifamily density near ward borders
2. $\sim$2% higher rents on the more stringent side of the border
3. Effect appears to operate through reduced supply

Related Literature

Time-to-Build, Frictions, & Supply Elasticity

- Development lags distort investment and misallocate resources (Budish et al., 2015; Hsieh and Moretti, 2019)
- Permitting lags and "process" are primary drivers of inelasticity (Paciorek, 2013; Gyourko et al., 2021; Molloy et al., 2022; Soltas and Gruber, 2026)

Regulatory Stringency & Discretion

- Stringency can increase the "option value of waiting" (Cunningham, 2006; Bloom, 2009)
- Discretionary review and local actors amplify delays (Fischel, 2009; Ortalo-Magné and Prat, 2014; Einstein et al., 2019; Khan, 2021; Duranton and Puga, 2023)

Border Discontinuity Designs

- School valuation (Black, 1999; Bayer et al., 2007)
- Spillovers and supply constraints (Turner et al., 2014; Diamond et al., 2019; Kulka et al., 2022)

A "Time-to-Build" Model of Regulatory Stringency

Key idea: Regulatory environment $\rightarrow$ effective delay $T(s)$ $\rightarrow$ discounts developer revenue

Developer's problem. A project chooses density $q \geq 0$ with cost $C(q) = F + cq + \frac{\gamma}{2}q^2$.

Revenue arrives upon completion, discounted by regulatory delay:

$$\max_{q \geq 0} \quad e^{-rT(s)}\, p\, q \; - \; C(q)$$

where r = discount rate, $T(s)$ = time-to-build under regulatory environment s

Optimal density (interior solution):

$$q^*(T, p) = \frac{e^{-rT(s)}\, p - c}{\gamma} \qquad \text{if } e^{-rT(s)} p > c, \text{ else } q^* = 0$$

Sufficient statistic: $rT(s)$ summarizes the regulatory wedge — captures delay, redesign, negotiation, financing frictions, and regulatory stringency

Testable Predictions

Main comparative static: $\partial q^* / \partial T(s) < 0$, which implies:

1. **Less density** — completed projects have fewer units → Density border-pair FE
2. **Higher rents** — lower aggregate supply $Q = \sum_i q_i^*$, $p'(Q) < 0$ → Rental border-pair FE

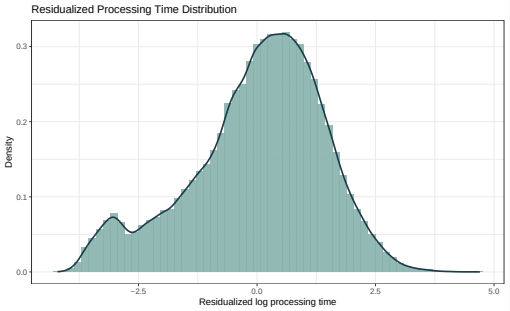
Mapping to data:

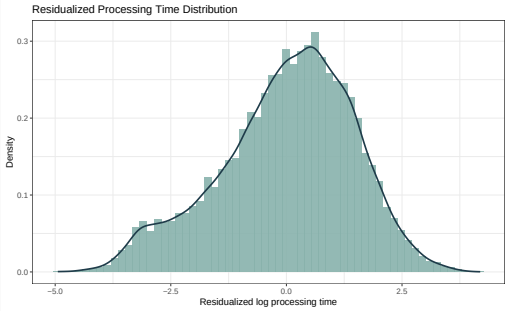
- $T(s)$ proxied by observed permit-to-completion times
- Regulatory stringency index $\approx$ alderman-specific component of $T(s)$
- Cross-boundary variation: different aldermen $\Rightarrow$ discrete changes in $T(s)$

The Challenge: Measuring Regulatory Stringency

- Alderman preferences/frictions are latent: need to infer from observed behavior
- **Idea**: Use permit processing times as a **proxy** for regulatory stringency and variation in time to build
  - Data: Universe of Permits in Chicago (2006-2025) from Department of Buildings
  - Longer average processing time $\rightarrow$ more back-and-forth, more frictions $\implies$ more stringent alderman
- Focus on **high-discretion permits**: new construction, renovation, demolition

Do permits Matter?





Stringency Index Construction

1. **Residualize** processing times on permit-level characteristics and a rich set of fixed effects (permit type, review type, and year $\times$ month)
   - Controls for baseline differences across wards (income, demographics, etc.)
2. **Estimate alderman fixed effects** on residualized processing times

$$\widehat{\epsilon}_{it} = \mu + \delta_a + \eta_{it}$$

3. **Empirical Bayes shrinkage** to reduce noise (Kline et al., 2022, 2024) then standardize to mean 0, SD 1

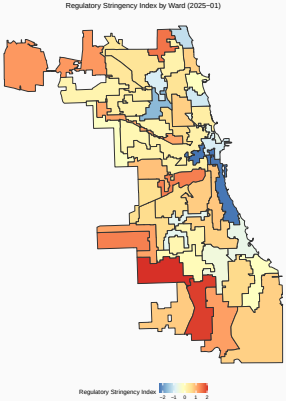
Interpretation: High score = alderman associated with longer processing times

Caveat: This is a *proxy*: likely understates true variation in bureaucratic frictions

- Misses projects that never apply or are discouraged early

Alderman Regulatory Stringency Index

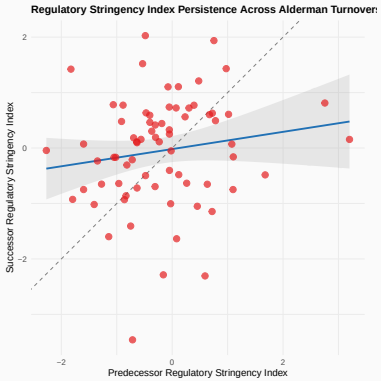
	Log Processing Time (1)
Median HH Income ($10k)	-0.001
	(0.007)
Share Black	0.049
	(0.076)
Share Hispanic	0.108**
	(0.054)
Share White	0.414***
	(0.084)
Homeownership Rate	0.130*
	(0.070)
Share Bachelor's+	-0.186***
	(0.070)
Population (10k)	-0.036***
	(0.013)
Dist. to CBD (km)	-0.013***
	(0.002)
Dist. to Lake (km)	0.0008
	(0.002)
CTA Stations (800m)	-0.019***
	(0.003)
Lag Permits	-0.001***
	(0.0004)
Observations	139,493
R^2	0.10968
Review Type FE	✓
Permit Type FE	✓
Year × Month FE	✓



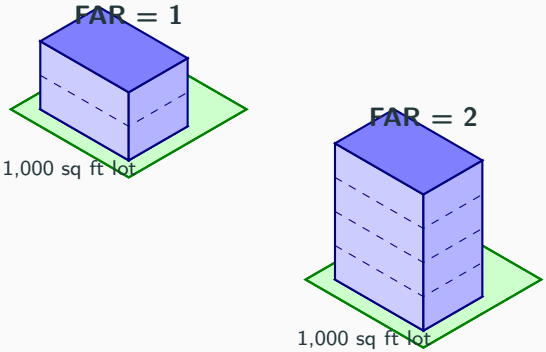
▸ Final Stringency Scores

Validation Checks: Permits and Within-Ward Correlation

Outcome (Ward-Month)	Coef. on Stringency Index
N permits (all high-discretion)	-9.48^{***}
	(0.36)
Share with any unit change	0.0081^{***}
	(0.0008)
Units reduced per permit (text)	0.0066^{***}
	(0.0025)
Ward-month observations	10,753



Density Outcomes: Floor Area Ratio (FAR)



$$\text{FAR} = \frac{\text{Total Building Floor Area}}{\text{Lot Area}}$$

FAR limits development intensity

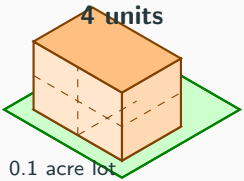
Same lot, different allowed density:

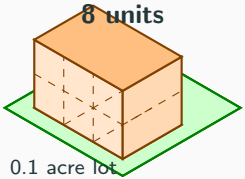
- FAR = 1: Can build 1,000 sq ft
- FAR = 2: Can build 2,000 sq ft

Chicago zoning examples:

Zone	Max FAR	Typical Use
RS-3	0.9	Single-family
RT-4	1.2	Two-flat
RM-5	2.0	Small apartment
RM-6	4.4	Mid-rise

Density Outcomes: Dwelling Units Per Acre (DUPAC)





DUPAC: 40 vs. 80 units/acre (same building volume)

DUPAC captures population density

$$\text{DUPAC} = \frac{\#\ \text{Dwelling Units}}{\text{Lot Acreage}}$$

Why both measures?

- Same FAR, different DUPAC
- **Developers** care about FAR (total rentable space)
- **Neighbors** care about DUPAC (congestion, parking, schools)

Data: New Multifamily Construction

Source: Cook County Assessor parcel data, 2006-2025

Sample: Multifamily buildings

- **Single family**: Majority of construction but less variation in density

Why multifamily matters:

- Only ∼19% of new construction but a disproportionate share of total housing supply

Summary statistics:

	All	MF Only
Avg FAR	1.22	2.32
Avg Units	6.65	31.3
Avg Stories	2.27	2.75
Avg DUPAC	28.5	74.5
N	13,236	2,469

Identification Strategy: Border Discontinuity

Idea: Compare new multifamily construction on opposite sides of ward boundary segments

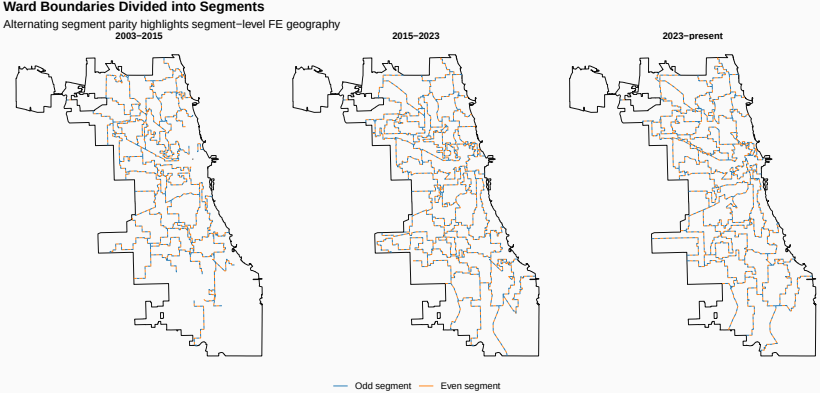
- Focus on parcels **within 500 feet** of a segment
- Fixed effects: zoning code, boundary segment, and year

Identifying variation: Within the same boundary segment, controlling for zoning code and citywide year fixed effects, does the more stringent side have less dense new multifamily construction?

Two specifications:

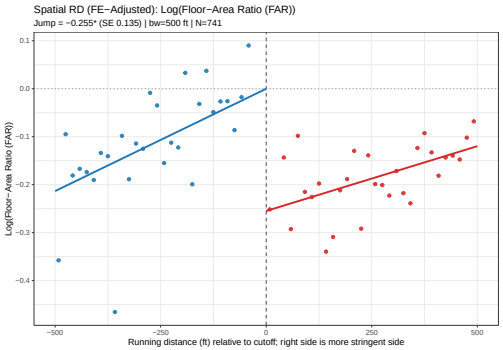
- **Visual RD**: FE-adjusted discontinuity plots
- **Regressions**: FE regressions using continuous variation in the stringency index

Border Segments

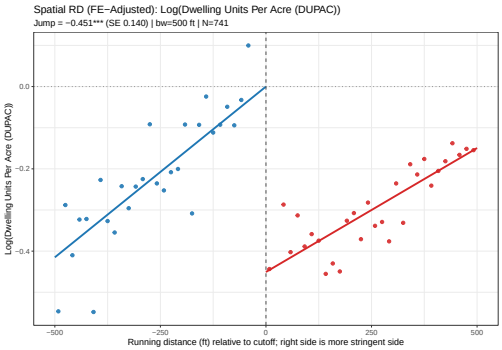


Main Results: FE-Adjusted Spatial RD at Ward Borders

Log(FAR)



Log(DUPAC)



Notes: Multifamily sample, 500 ft bandwidth, zoning + segment + year FE, segment-clustered SEs.

▸ All Units ▸ Median Split ▸ Donut RD ▸ Boundary Placebo

Main Results: Border FE Regressions

Specification: $\ln(Y_i) = \beta \cdot \text{Stringency}_i + \alpha_{\text{zone}} + \alpha_{\text{segment}} + \alpha_{\text{year}} + \varepsilon_i$

	ln(FAR) (1)	ln(DUPAC) (2)	ln(Units) (3)
Stringency Index	-0.10***	-0.07**	-0.001
	(0.04)	(0.04)	(0.04)
Observations	741	741	741
Dep. Var. Mean	2.48	81.90	31.97
Ward Pairs	98	98	98
Zoning Code FE	✓	✓	✓
Segment FE	✓	✓	✓
Year FE	✓	✓	✓

Interpretation: A 1 SD ↑ in stringency is associated with $7 - 10\%$ lower multifamily density.

▸ Pruned Sample ▸ Zone + Ward-Pair + Year FE ▸ Other Bandwidths

From Density to Rents

So far: More stringent aldermen $\rightarrow$ lower multifamily density

Next question: Does this supply constraint feed into higher housing costs?

Approach: Apply the same border-pair FE design to rental listings

- Data: RentHub rental listings, 2014-2022
- 2.67 million listing observations, 141 ward pairs, BW = 500 ft
- Segment $\times$ year-month FE; SEs clustered at segment level

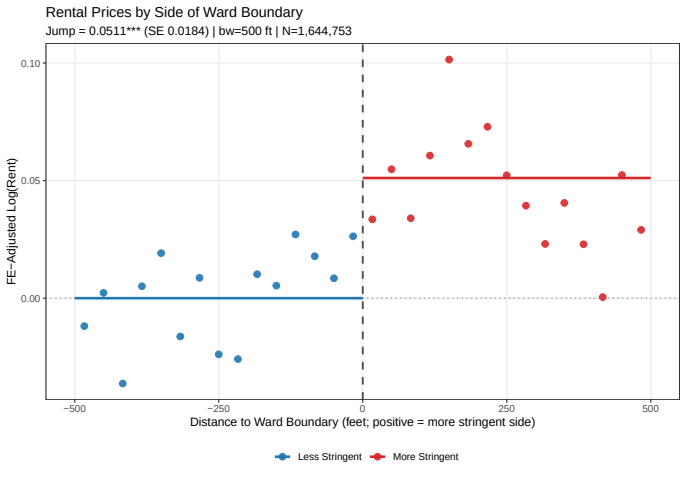
Effect on Rental Prices

	(1)	(2)
Stringency Index	0.034**	0.029***
	(0.016)	(0.010)
Observations	2,664,657	1,644,753
Dep. Var. Mean	2521	2799
Ward Pairs	140	139
Hedonic Controls		✓
FE Structure	Segment × Year-Month FE	Segment × Year-Month FE
Cluster Level	Segment	Segment

⇒ A 1 SD ↑ in stringency ⇒ 3.4% higher rents (no controls) or 2.9% (with controls).

▸ Permutation Test

Spatial RD: Rental Prices at Ward Borders



Home Sale Prices

	(1)	(2)
Stringency Index	0.011*	0.010*
	(0.006)	(0.005)
Observations	55,043	54,986
Dep. Var. Mean	395168	394905
Ward Pairs	156	156
Hedonic Controls		✓
Segment × Year-Quarter FE	✓	✓
Cluster Level	Segment	Segment

Ruling Out Quality Sorting

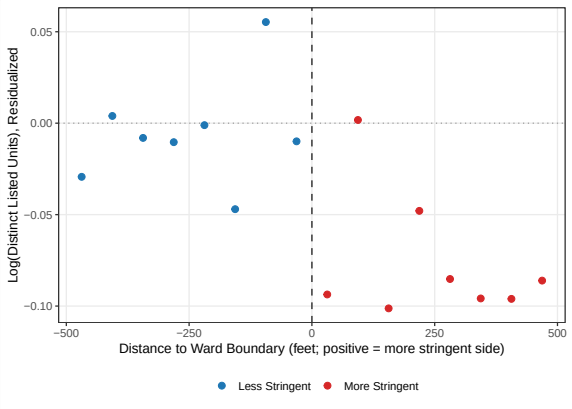
Concern: Rent premium could reflect *better* units on the more stringent side

Test: Same border-pair FE on unit characteristics (BW = 500 ft, 2014–2022)

	Sqft	Beds	Baths	Multi-Family	Laundry	Gym
More Stringent Side	-0.9245	-0.0090	-0.0088	0.0181***	0.0269	0.0117
	(29.4958)	(0.0560)	(0.0414)	(0.0069)	(0.0232)	(0.0194)
Observations	2,037,444	2,664,657	2,660,323	2,664,657	2,664,657	2,664,657
Dep. Var. Mean	894.81	1.38	1.30	0.76	0.59	0.46
Segment × Year-Month FE	✓	✓	✓	✓	✓	✓
Clustered by Segment	✓	✓	✓	✓	✓	✓

⇒ Core size/bed/bath margins are close to zero; composition shifts are small.
Quality sorting is unlikely to explain the full rent premium at ward boundaries.

Supply Channel: Fewer Available Rentals



	Distinct Units
Stringency (1 SD)	-0.0929*
	(0.0543)
Observations	94,546
Estimator	PPML
FE Spec.	Seg. × Year-Month

Outcome: Distinct listed units (unit-proxy key), ward-pair × side × month
Interpretation: 1 SD ↑ in stringency ⇒ ∼9% fewer listed units

Discussion & Policy Implications

Core problem: Regulatory stringency and discretion raise the cost of housing development and lead to lower density

- Longer permitting times + more uncertainty → higher carrying costs for developers
- Fewer projects initiated → constrained rental supply → higher rents
- Effects stronger in the rental market, but appears to spill over to housing market as well

Policy alternatives to reduce regulatory friction:

1. **Reduce Individual Discretion**: Remove aldermanic privilege and ability to change zoning or delay permits when development is by-right
2. **Transfer decision-making authority**: Shift zoning decisions to the citywide level (i.e. mayor, zoning board) to reduce fragmentation and local capture

Conclusion

Summary of findings:

1. More stringent aldermen $\Rightarrow$ lower multifamily density at ward borders by $\sim$7-10%
2. $\sim$3% higher rents which appears to be driven by reduced supply

Implications:

- Regulatory stringency in decentralized zoning lowers supply and raises rents
- Within-city fragmentation matters, not just across municipalities
- Reforms should target decision-making authority and discretion, not just by-right zoning reforms

Thank you!

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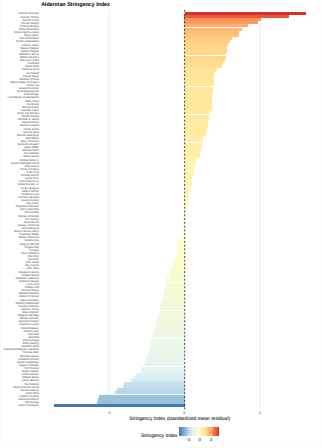
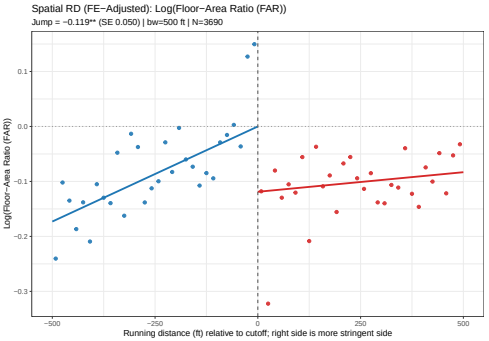
Alderman Regulatory Stringency Index



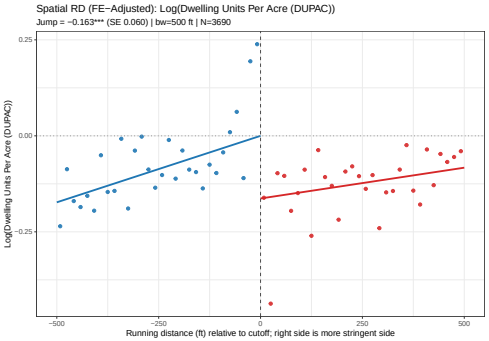
Back

FE-Adjusted Spatial RD: All Units

Log(FAR)



Log(DUPAC)



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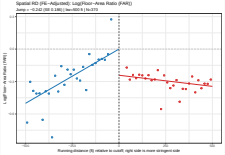
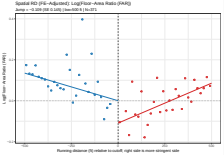
FE-Adjusted Spatial RD: Below/Above Median Gap

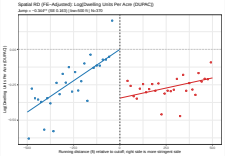
FAR: Below Median



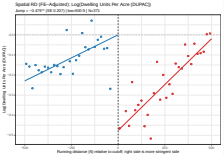
FAR: Above Median



DUPAC: Below Median



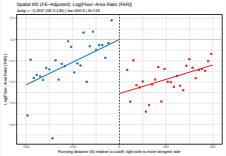
DUPAC: Above Median



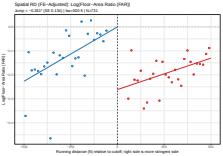
Back

FE-Adjusted Spatial RD: Donut RD

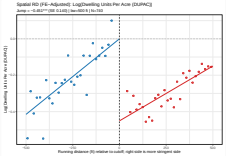
FAR: Donut 25 ft



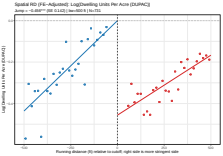
FAR: Donut 50 ft



DUPAC: Donut 25 ft



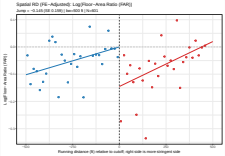
DUPAC: Donut 50 ft



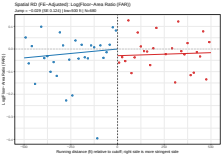
Back

FE-Adjusted Spatial RD: Boundary Placebo (FAR)

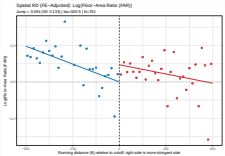
Shift -500 ft



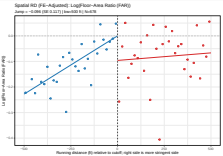
Shift -250 ft



Shift +250 ft



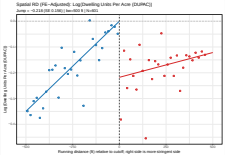
Shift +500 ft



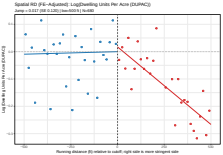
DUPAC Placebo
Back

FE-Adjusted Spatial RD: Boundary Placebo (DUPAC)

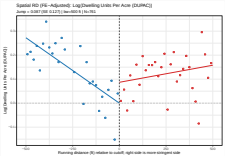
Shift -500 ft



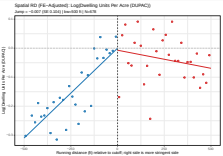
Shift -250 ft



Shift +250 ft



Shift +500 ft



Back

Border FE Robustness: Pruned Sample

	ln(FAR) (1)	ln(DUPAC) (2)	ln(Units) (3)
Stringency Index	-0.12**	-0.06	0.004
	(0.06)	(0.06)	(0.06)
Observations	622	622	622
Dep. Var. Mean	2.19	74.42	26.38
Ward Pairs	82	82	82
Zoning Code FE	✓	✓	✓
Segment FE	✓	✓	✓
Year FE	✓	✓	✓

Back

Border FE Robustness: Zone + Ward-Pair + Year FE

	ln(FAR) (1)	ln(DUPAC) (2)	ln(Units) (3)
Stringency Index	-0.05*	-0.06*	0.009
	(0.03)	(0.03)	(0.03)
Observations	741	741	741
Dep. Var. Mean	2.48	81.90	31.97
Ward Pairs	98	98	98
Zoning Code FE	✓	✓	✓
Ward-Pair FE	✓	✓	✓
Year FE	✓	✓	✓

Back

Border FE Robustness: Bandwidth = 250 ft

	ln(FAR) (1)	ln(DUPAC) (2)	ln(Units) (3)
Stringency Index	-0.08	-0.04	0.06
	(0.08)	(0.07)	(0.04)
Observations	430	430	430
Dep. Var. Mean	2.67	89.20	35.66
Ward Pairs	84	84	84
Zoning Code FE	✓	✓	✓
Segment FE	✓	✓	✓
Year FE	✓	✓	✓

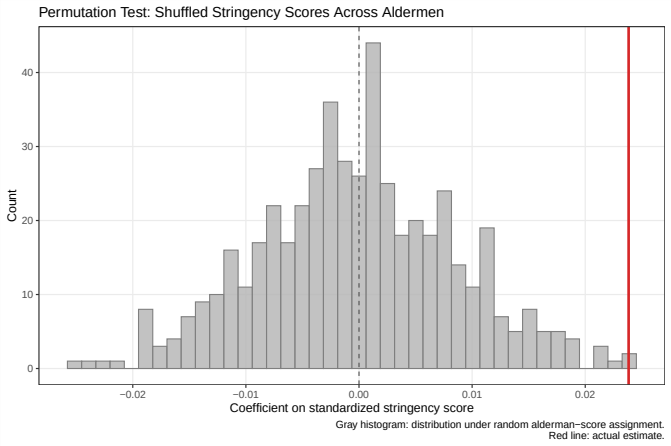
1000 ft
Back

Border FE Robustness: Bandwidth = 1000 ft

	ln(FAR) (1)	ln(DUPAC) (2)	ln(Units) (3)
Stringency Index	-0.06**	-0.06***	-0.04
	(0.02)	(0.02)	(0.04)
Observations	1,279	1,279	1,279
Dep. Var. Mean	2.31	75.53	29.11
Ward Pairs	111	111	111
Zoning Code FE	✓	✓	✓
Segment FE	✓	✓	✓
Year FE	✓	✓	✓

250 ft
Back

Permutation Test: Rental FE (Controls)



2014–2022 sample, 500 ft bandwidth, controls included.

Permutation p-values: two-sided = 0.006; one-sided = 0.004.